

MID SEM EXAMINATION MAY'2022
Department of Mathematics, NITPatna
Course Code: MA24101 [Probability and Statistics]

Branch: CSE(2nd Sem)

Answer any four questions

Marks:30

Time : 2 hours

- 1 (a) An urn contains 4 white and 6 black balls. Two balls are successively drawn from the urn without replacement of the first ball. If the first ball is seen to be white, what is the probability that the 2nd ball is also white.
- (b) There are two identical urns containing respectively 4 white, 3 red balls and 3 white, 7 balls. An urn is chosen at random and a ball is drawn from it. Find the probability that the ball is white. If the ball drawn is white, what is the probability that it is from the first urn? $\frac{4}{10}$
- 2(a) There are 3 true coins and 1 false coin with 'head' on both sides. A coin is chosen at random and tossed 4 times. If 'head' occurs all the 4 times, what is the probability that the false coin has been chosen and used?
- (b) For a binary, communication channel, the probability that a transmitted '0' is received as a '0' is 0.95, if the probability that a '0' is transmitted is 0.4, find the probability that
- (i) a '1' is received and (ii) a '1' was transmitted given that a '1' was received.
- 3(a) A consignment of 8 similar Mobile to retail out-let contain 3 that are defective. If a firm makes a random purchase of 2 of these Mobiles. Find the probability distribution for the number of defectives.
- (b) Suppose that the error in the reaction temperature ($^{\circ}\text{C}$), for a controlled laboratory experiment is continuous random variable X having the probability density functions

$$F[X = x] = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Verify that of total probability is unity (ii) Find $P(0 \leq X, 1)$

- 4(a) If the mean of a binomial distribution is 3 and the variance is $3/2$, find the probability of obtaining at most 3 success. $2/32$
- (b) The Probability that a certain kind of component will survive a given shock test is $3/4$. Find the probability that exactly two of the next four components tested will survive. $\frac{27}{128}$
- 5(a) In a certain factory turning razor blades, there is a small chance, $1/500$ for any blade to be defective. The blades are in a packets of 10. Use **Poisson** distribution to calculate the approximate number of packets containing (i) no defective (ii) one defective (iii) two defective blades respectively in one consignment of 10000 packets. (Given $e^{-0.02} = 0.9802$) 9802

- (b) The average number of radioactive particles through a counter during 1 milli second in a laboratory experiment is 4. What is the probability that 6 particles enter the counter in a given millisecond.

- 6 The length of bolts produced by a machine is normally distributed with mean 4 and s.d 0.5. A bolt is defective if its length does not lie in the interval (3.8, 4.3). Find the percentage of defective bolts produced by the machine. (Given $\frac{1}{\sqrt{\pi}} \int_{-\infty}^{0.6} e^{-\frac{t^2}{2}} dt = 0.7257$; $\frac{1}{\sqrt{\pi}} \int_{-\infty}^{0.4} e^{-\frac{t^2}{2}} dt = 0.6554$) 0.6029

OR

- 6 If weekly wages of 10000 workers in a factory follows normal distribution with mean S.D. Rs.70 and Rs.5 respectively, find the expected number of workers whose weekly wages are (i) between Rs. 66 and Rs. 72 (ii) less than Rs. 66 (iii) more than Rs. 72.

(Given that $\frac{1}{\sqrt{2\pi}} \int_0^z e^{-t^2/2} dt = 0.1554$ and 0.2881 according as $z=0.4$ and $z=0.8$)